

DBT Interview Question with Answer

Q.1. Why do we need a database? Can't we store data in files?

Answer:

If I have to explain why we need a database, I would say that while we can store data in files, it is not efficient for real-world applications.

In file systems:

- Data redundancy can occur
- Searching data is slow
- It is difficult to maintain relationships between data
- There is no proper security or concurrency control

A database like MySQL solves these problems by:

- Organizing data in a structured format (tables)
- Providing fast querying using SQL
- Supporting multiple users at the same time
- Ensuring data consistency and integrity

So in real-world applications like banking or e-commerce, databases are essential for managing large and related data efficiently.

Q.2. What is a database and how is data organized in MySQL?

Answer:

A database is a structured collection of data that is stored in a way that makes it easy to access, manage, and update. It helps us organize large amounts of information efficiently instead of keeping it in scattered files.

In MySQL, which is a relational database management system, data is organized in the form of tables. A table is similar to an Excel sheet where data is arranged in rows and columns.

- **Tables** → Used to store data in an organized format
- **Rows** → Each row represents a single record (for example, one student's details)
- **Columns** → Each column represents a specific attribute or field (like name, age, marks)

For example, in a **Student table**:

- Columns: id, name, marks
- Rows:
 - (1, Rahul, 85)

- (2, Sneha, 90)

This structured format helps in:

- Easy data retrieval using queries
 - Efficient data storage
 - Maintaining consistency and avoiding redundancy
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Q.3. What is a Primary Key and why is it important?

Answer:

A Primary Key is a column or a combination of columns in a table that uniquely identifies each record in that table. It ensures that no two rows have the same value in that column.

For example, in a student table, student_id can be used as a primary key because each student has a unique ID.

Important properties of a Primary Key:

- It must contain **unique values**
- It **cannot be NULL** (every record must have a value)
- There can be only **one primary key** per table

Why it is important:

- It prevents duplicate records in the table
- It helps in identifying each row uniquely
- It is used to establish relationships with other tables (through foreign keys)

In real-world applications, primary keys are very important for maintaining data accuracy and ensuring each record can be uniquely accessed.

Q.4. What is a Foreign Key and why do we use it?

Answer:

A Foreign Key is a column (or set of columns) in one table that refers to the Primary Key of another table. It is mainly used to create a connection between two tables.

For example:

- A **Student table** has student_id as Primary Key
- An **Orders table** may have student_id as a Foreign Key

This means each order is linked to a valid student.

Why we use Foreign Keys:

- To create relationships between tables
- To maintain **referential integrity** (valid connections between data)
- To avoid invalid or inconsistent data

In real-world systems:

- You cannot add an order for a student that does not exist
 - It ensures data remains accurate and meaningful
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Q.5. Difference between Primary Key and Foreign Key

Answer:

The main difference between a Primary Key and a Foreign Key is based on their purpose.

- A **Primary Key** is used to uniquely identify each record within its own table
- A **Foreign Key** is used to create a relationship between two tables

Other differences:

- Primary Key:
 - Must be **unique**
 - Cannot be **NULL**
 - Only one per table
- Foreign Key:
 - Can have **duplicate values**
 - Can be **NULL** (depending on constraints)
 - Multiple foreign keys can exist in a table

Simple understanding:

- Primary Key → Identity of a record
 - Foreign Key → Connection between tables
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Q.6. What are constraints in MySQL and why are they used?

Answer:

Constraints are rules applied to table columns in MySQL to ensure the accuracy, reliability, and integrity of the data stored in the database.

They help control the type of data that can be inserted into a table.

Common types of constraints:

- **PRIMARY KEY** → Uniquely identifies records
- **FOREIGN KEY** → Maintains relationships between tables
- **NOT NULL** → Ensures a column cannot have empty values

- **UNIQUE** → Ensures all values in a column are different

Why constraints are used:

- To prevent invalid or incorrect data entry
- To maintain consistency across the database
- To enforce business rules (e.g., no duplicate emails)

In real-world applications, constraints make sure the database remains reliable and reduces chances of errors.

Q.7. What is NOT NULL and UNIQUE constraint?

Answer:

These are two important constraints used in MySQL to maintain data quality.

NOT NULL Constraint:

- Ensures that a column cannot have a NULL (empty) value
- Every record must contain a value in that column

Example:

If name is NOT NULL, then every student must have a name.

UNIQUE Constraint:

- Ensures that all values in a column are different
- No duplicate values are allowed

Example:

An email field is usually marked as UNIQUE because each user should have a different email.

Difference between them:

- NOT NULL → Prevents empty values
- UNIQUE → Prevents duplicate values

In real-world usage:

- NOT NULL ensures required data is always present
- UNIQUE ensures important fields like email, phone number remain distinct